10-Packet Implementation Strategy

Portable reference for a 5-lakh concurrent-attempt target

Full strategy • measurable gates • audit baseline • context restore

STATUS: STRATEGY SNAPSHOT — IMPLEMENTATION NOT STARTED

Packet 1 is approved to begin after this PDF is delivered. This document does not certify capacity, fix audit findings, provision infrastructure, rotate staff credentials, or approve an exam for publication.

Non-negotiable: phone-only student entry, matte-maroon UI, exact question–solution pairing, offline answer recovery and honest evidence-backed progress.

Version: 1.0
Generated: 2026-10-12T13:08:44+00:00
Included: all 10 packets, SLO/workload contract, operator gates, full audit/errata, 62 requirement rows and 60 analysis records.

Keep this file with the five original specification PDFs. Upload it in a future session and ask the agent to read “Context restore” before continuing.

Context restore — read before implementation

Snapshot v1.0: strategy reference, not an implementation or capacity certificate.

Latest user decision

The user approved reading the strategy/audit/rulebook and implementing **Packet 1 only**. Their immediate priority is: “Give me first 10 packets pdf so that whenever you forget i will send you back”. This export is delivered before Packet 1 application changes.

- **All 10 implementation packets: NOT STARTED at this snapshot.** Packet 1 is approved to begin after this PDF is delivered; no packet exit gate has passed.
- Student entry must remain **phone-number only, without OTP/password**. It is not verified phone ownership or verified candidate identity.
- The UI must remain **matte maroon (#5C0025)**. Strengthen staff authentication separately.
- Investigate the exposed staff credential safely; obtain **separate user confirmation before rotating the persisted credential or invalidating staff sessions**. No rotation/revocation happened during this export.
- Do not claim support for **500,000 concurrently active exam attempts** until all required qualification gates pass for the declared deployment.

How to resume in a later session

1. Read this handoff, the complete 10 packets, workload contract and qualification appendix. Treat them as a versioned snapshot, not evidence that code was implemented.
2. Re-read the current repository plan and compare source hashes in this PDF against current documents. Newer verified evidence and explicit user decisions supersede this snapshot; do not revert completed work blindly.
3. Review the audit, errata and current runtime. Follow Packet 1's actual scope: authoritative requirements, API/data contracts, safe regression baseline, authorized containment and evidence registries.
4. Keep permanent defect repairs in their assigned packets: A05/A06/S01 in 04; A02/A03/A04 in 05; A01 in 06; F01/F02/F03 in 07. Do not call a failing reproduction a fix.
5. Confirm critical security operations separately, run independent tests for any code changes, and update the plan/handoff after each packet.

Repository and runtime boundaries

- Backend: FastAPI, domain modules under `/app/backend/tejas/`.
- Frontend: React, Shadcn and Tailwind under `/app/frontend/src/`.
- Database: current pilot MongoDB; the audit describes a shared standalone writer, not independent live cells.
- Strategy: `/app/docs/strategy/TEN_PACKET_STRATEGY.md`, `SLO_AND_WORKLOAD.md`, `QUALIFICATION_AND_HANDOFF.md`.
- Audit: `/app/docs/audit/AUDIT_REPORT.md`, `REQUIREMENTS_MATRIX.md`, `RULEBOOK_ERRATA.md`.
- Rulebook: `/app/docs/RULEBOOK_ARCHITECTURE_V5.md`; read with errata. Its legacy “40 analyses” statement is wrong: **all 60 are required**.

- All application APIs remain under `/api`. Never rewrite the existing `REACT_APP_BACKEND_URL` or `MONGO_URL` values.
- Infrastructure provisioning, DB administration, SSL, CI/CD and production deployments require authorized operators, not silent builder changes.

Evidence limitations and known gaps

The existing strategy still says “proposed for approval” because its original text is preserved. The latest user decision above approves Packet 1 sequencing; it does **not** sign off provider choices, exact proposed SLOs, budgets, academic publication or production readiness.

The strategy references `coverage_manifest.json` and `COVERAGE_MATRIX.md`, but those files were **not found in the current docs inventory**. They are not supplied as verified coverage. Packet 1 must resolve that gap and expand the 62 audit rows into atomic mandatory-clause traceability. Page counts and an inventory are not proof of complete requirements coverage.

Existing audit observations and test counts are historical, not re-run during this PDF export. Services were running at export preparation; that check is not functional or load qualification. The original paper remains unapproved according to the audit; do not publish it to make testing easier.

This PDF includes the full three strategy documents, full audit and errata, the 62-row requirements matrix, the 60-analysis status inventory, and source identities/hashes. The original five specification PDFs (177 pages) are **not embedded or reproduced**. Keep them with this reference when complete clause-level authority is needed. No raw test reports, source-document credentials, private URLs or live secrets are included.

Status to carry forward

- Working practice MVP and matte-maroon/phone-only choices exist; correctness and security gaps remain open as documented.
- Leaked staff credential: **rotation/revocation still pending**.
- Independent worker fleet, PostgreSQL live cells, full analytics and load/fault/restore qualification: **not delivered by this document**.
- Next development action: **execute Packet 1 only**, after delivery of this reference.

Contents

Click an entry or use the PDF bookmarks. Appendices are reference material, not extra implementation packets.

Context restore — read before implementation . . . 2

The 10-packet strategy — full source . . . 5

Packet 01 — Authoritative Rules, Safety & Regression Baseline . . . 7

Packet 02 — Database & Live-Cell Foundation . . . 8

Packet 03 — Identity, Permissions & Locked Trust . . . 9

Packet 04 — Durable Workers & Safe Processing . . . 10

Packet 05 — Content Sourcing, Catalogue & Publication . . . 11

Packet 06 — Exam Engines, Durable Sessions & Exact Scoring . . . 12

Packet 07 — Frontend, Offline Recovery & Edge Experience . . . 13

Packet 08 — All 60 Analyses, Cohorts & Study Plans . . . 14

Packet 09 — Results, Released Solutions & PDF Reports . . . 15

Packet 10 — Full-Scale Qualification, Recovery & Operations . . . 16

Appendix A — Workload and SLO contract . . . 18

Appendix B — Qualification and operator handoff . . . 23

Appendix C — Full architecture audit . . . 28

Appendix C.1 — Rulebook errata . . . 35

Appendix D — All 62 audited requirement rows . . . 37

Appendix E — All 60 analysis records . . . 46

Appendix F — Traceability and source hashes . . . 58

Appendix G — Status and next development action . . . 60

The 10-packet strategy — full source

Status: STRATEGY ONLY — proposed for approval. No packet implementation has started.

The contract

Build for **500,000 concurrently active exam attempts**, not registered users or idle connections. Preserve **matte maroon**, accessible light/dark modes, and **phone-number-only student entry without OTP/password**. Strengthen staff security separately. A phone profile is not verified human identity; do not claim proctored identity assurance or verified-person cohort counts under this choice.

Authority: all five supplied PDFs (177 pages), their verified source manifest, explicit user overrides, and the audit. The internal “40 analyses minimum” is wrong: **all 60 analysis contracts are required**. Current app/source state remains unchanged; original75-question paper is review-only and QA sets remain withdrawn.

No launch/capacity claim while any required correctness, security, recovery or load gate is failed or unqualified. “No lagging” means passing `SLO_AND_WORKLOAD.md`, not promising zero delay on every network or through unbounded disasters.

Architecture we will build and qualify

- **Edge:** static frontend and authorized candidate-asset distribution; viewer authorization before protected cache hits; no source/key distribution.
- **Control/trust:** separate identity, entitlements, signed academic decisions, staff grants, catalogue and admission preparation. Admitted writes must not synchronously depend on a global allocator or analytics service.
- **Live cells:** reference layout20 independently qualified cells × at most25,000 active attempts, each with independent transactional writer/HA boundary, bounded API pools, local authority, outbox and reserved terminal capacity.20pods/databases sharing one writer do not qualify.
- **Batch/analytics:** autonomous, isolated extraction, grading, report, regrade and aggregation execution. Redis/caches/queues are not the source of acknowledged response truth.
- **Operations:** protected evidence, metrics, quotas, keys, backup/restore, incident control, deployment manifests and release decisions.

The reference live authority is **PostgreSQL with a specifically verified synchronous durability/failover configuration** because the guide specifies relational lock-time semantics. Final managed service, engine version, regions and sizing require operator selection and proof. Mongo may remain a segregated legacy/archive store during migration, not an assumed equivalent live-cell authority. Do not rewrite historical attempt truth or invent a cross-store transaction.

Execution discipline

1. The packets below are dependency ordered. Each has implementation deliverables and explicit exit evidence. An implementation gate is not a500k qualification gate.
2. Packet01 first freezes atomic requirements/contracts and isolates existing red regressions. Their permanent owners are named below; containment is not called a fix.
3. Every later fix must pass its original failing sequence through a **new testing-agent report** plus regression checks. Tests must call real implementation; fixtures/mocks are labelled; a test reproducing a bug does not close it.

4. No production/student traffic uses an affected capability until its defect gates close. No automatic publication or simulated academic approval.
5. Observability, security, migration and cost evidence are added from the start, not postponed to Packet10.
6. Application engineers own code/contracts/tests. Authorized cloud/SRE/DBA/security operators own provisioning, DB-server configuration, IaC, CI/CD, SSL, capacity reservations and production deployments. Those infrastructure actions are outside this builder session's operational scope.
7. Existing preview `REACT_APP_BACKEND_URL` and `MONGO_URL` must not be rewritten. Future provider configuration is additive and operator-managed. All application APIs remain under `/api`; guide `/v1`/internal routes receive explicit compatible aliases, not broken ingress paths.

Packet 01 — Authoritative Rules, Safety & Regression Baseline

Depends on: strategy approval. **Owners:** product/academic owner, application lead, QA, security operator.

Build/deliver

- Atomize every MUST/acceptance clause in the five PDFs into versioned requirements with owner, API/data contract, dependency, acceptance test and evidence location. The62audit rows and page index are starting coverage, not proof of complete atomic traceability.
- Correct rulebook errata; preserve all60analyses, all13operational WP gates and all12OPS-R release blockers. Freeze workload, SLO measurement endpoints, critical windows, supported device/network profiles and failure envelope.
- Publish the contract-first API/permission catalogue: strict input/output DTOs, field allowlists, canonical body hashing, typed errors, idempotency scope, ETags/If-Match, status/Location/Retry-After, payload budgets, UTC/monotonic clock units and exact marks.
- Establish repository/domain boundaries, real-function regression fixtures, independent expected ledgers, source integrity/SBOM baselines, secret-safe test output, isolated test data access and a requirement→test→artifact registry. Test repositories must fail closed if pointed at production namespaces.
- With explicit operator authorization, rotate the exposed persisted staff credential and revoke affected sessions. Merely updating a bootstrap environment value is insufficient. Contain unsafe upload/publication/job paths while their owning packets repair them; no public exam launch.
- Freeze source/role/retention/solution-release policies and the decision register in `QUALIFICATION_AND_HANDOFF.md`. Define cost/quotas and minimal metric schemas now.

Interfaces/artifacts: versioned OpenAPI/operation catalogue, data ownership map, requirement registry, workload/SLO manifest, regression corpus, incident record, release-blocker register.

Exit gate: owner-approved contracts and failure scope; zero orphan mandatory clauses in the detailed registry; all current findings have a red regression or explicitly unverified reproduction task; safe test isolation and credential containment evidence. **No defect is marked fixed merely because its test exists.**

Packet 02 — Database & Live-Cell Foundation

Depends on:01. **Owners:** application/data engineers; DBA/SRE/cloud operators for provisioning.

Build/deliver

- Provision/design one nonproduction reference cell and separate control/trust stores, then define reproducible fleet topology. Verify exact primary/standby acknowledgement and failover semantics; never silently downgrade durability when a standby disappears.
- Define UUID identities and constraints for reservations, attempt ownership, writer generations, command log/receipts, committed responses, drafts, phase transitions, submission receipts, grade inputs and transactional outbox. Control data owns source/version/review/grant/decision records; analytics owns derived snapshots only.
- Use SQL uniqueness/CAS/locks for truth, append-only version/history records, time/hash partitioning justified by query patterns, bounded connection pools/wait, appropriate indexes, archival/retention and measured WAL/fsync/IOPS/vacuum/storage budgets. Pool totals across replicas/workers must fit a fleet DB budget.
- Build cell-routing/admission primitives and signed home-cell routing metadata. Pre-distribute pinned schemas/authorized versions and verification material; critical writes avoid global user/catalogue lookups. Freeze specific authority/freshness behavior during control-plane outages.
- Migrate safely: map legacy UUIDs, pack hashes, versions, source refs and receipts; reconcile counts/checksums/scores; shadow-read/grade; expand before contracting schemas. Existing attempts stay on one authoritative backend until safely drained. No unfenced dual writers or blind dual-write migration.
- Establish protected telemetry, backup/key/object inventories, restore environment and signed release/IaC requirements with operators. Add cache and analytics stores only as nonauthoritative dependencies.

Interfaces/artifacts: repository interfaces, SQL schemas/migrations, data-class/retention map, HA/failover contract, migration/rollback evidence, per-cell connection and storage budgets, operator topology manifest.

Exit gate: schema/constraint and migration golden tests pass; canary data platform and operational ownership exist; selected durability semantics and controlled failover tests are recorded. **This is not yet a25k-cell or500k qualification.**

Packet 03 — Identity, Permissions & Locked Trust

Depends on:01–02. **Owners:** application/security engineers, identity/KMS operators, academic/legal owners.

Build/deliver

- Keep student phone-only entry, revocable secure sessions and profile disclosure. Bind authorization to tenant/profile/attempt/version; define abuse controls that work behind shared mobile NAT without blocking legitimate cohorts. Do not claim verified phone ownership or official candidate identity.
- Implement staff SSO/MFA, least-privilege capabilities, distinct author/reviewer/publisher/support/operator roles, session revocation and audited break-glass. Source/code/configuration/role-grant changes are protected operations too; “owner” is not a permanent bypass.
- Implement the guide's privileged step-up/PIN policy using protected credentials—not a shared PIN copied from the PDF. Issue one-use grants bound to actor, tenant, session assurance, resource/action, parent version, canonical diff/reason hashes and nonce; atomically consume with the protected operation.
- Separate encryption, signing and secrets purposes. Create canonical request/grant/result and manifest schemas, managed signing/verification, key rotation/survival, monotonic activation generations, independent approvals, supersession/revocation/correction lineage and current-authority checks. A valid historical signature is not proof that a release is still authorized.
- Establish separate workload identities for quarantine, extraction, draft editing, publication, scoring, delivery, signing and audit export. Viewer authorization and release/rights checks precede cache access. Define normal withdrawal versus emergency revocation for admitted attempts.
- Resolve privacy/retention/age-consent obligations, product telemetry opt-in/minimization, support access, exports/deletion and backup expiry. Under unverified phone entry, irreversible recovery/deletion policies require explicit approval; do not secretly add student passwords/OTP.

Interfaces/artifacts: permission matrix; identity/session/grant APIs; protected operation catalogue; signing/verifier services; key/retention/revocation runbooks; privacy and telemetry inventory.

Exit gate: cross-role/tenant/version access tests, expired/replayed/altered grant tests, independent approval tests and key-rotation/revocation tests pass; secrets never appear in fixtures/reports; student entry remains phone-only. Provider-backed controls must be real before a production gate passes.

Packet 04 — Durable Workers & Safe Processing

Depends on:01–03. **Owners:** backend/processing engineers; queue/compute/security operators.

Build/deliver

- Replace browser/cron-dependent dispatch with autonomous managed queue delivery and dedicated workers. Use cell/control outboxes, bounded inflight work, fair scheduling, backpressure, retry budgets, delayed retry, dead letters, poison isolation and clear operator recovery. Recurring repair/cleanup uses the platform-managed scheduler; it is not the primary grader dispatcher.
- Isolate grade, extraction/OCR, rendering, reports, regrade and analytics CPU/memory/connection/identity pools. Define per-tenant/cell/operation budgets; optional work cannot take admitted answer-write capacity.
- Close **A05**: queued work drains when capacity becomes free without polling or maintenance intervention; failed early entries cannot starve later work.
- Close **A06** at the destination: generation/token and required lease validity participate in the authoritative output transaction/CAS. Checking a lease in a different store before writing is not fencing. Immutable artifact candidates may exist, but only the currently authorized generation can publish the pointer.
- Close **S01** and triage dependency advisories: enforce total request, file, field, count and concurrency limits before expensive multipart parsing. Quarantine untrusted PDFs, sniff actual types, reject active/embedded content, and run parsers with restricted network/credentials/filesystem, bounded expanded resources and hard kill/retry limits.
- Implement job schemas and handlers' integration boundaries. Grade logic is completed in06, analytics in08, final reports in09; a generic worker fixture does not certify those later handlers.

Interfaces/artifacts: durable job APIs/receipts, outbox relay/consumers, destination fence protocol, sandbox profiles, resource/cost budgets, queue health/age metrics and recovery tools.

Exit gate: original A05/A06/S01 regressions pass independently;
worker-kill/lost-ack/duplicate/lease-handoff/poison/budget tests preserve truth and autonomous progress. No synchronous PDF/OCR work remains in interactive handlers. Physical isolation requires operator proof, not just process semaphores.

Packet 05 — Content Sourcing, Catalogue & Publication

Depends on:01–04. **Owners:** content/backend engineers; named academic reviewers/publishers and rights owner.

Build/deliver

- Build quarantine→extraction→review→locked truth→delivery pipelines for raw questions, solutions, separately supplied keys and combined papers. Record original bytes/hashes, rights evidence, source role, 1-based/0-based page conversion, anchors, language, OCR confidence and processing provenance. PDF/AI text is untrusted data, never execution instructions or approval.
- Implement question/solution/stimulus blocks, taxonomy, exact pairing cardinalities, duplicate detection, anomaly/key adjudication, formula/diagram review, accessible text, locality-aware previews and review diffs. Shared passages and multipart solutions retain stable identities.
- Close **A03** through expected-revision edits; close **A02** through one atomic publication transition and unique version identity; close **A04/A04b** through enum/date/schema validation and fail-closed rights/audience/expiry checks on every path.
- Separate `source-private`, `candidate-safe`, `server-key` and `embargoed-solution` artifacts physically/by workload authority. Pre-render/build them once at approved publication; do not parse source PDFs or repack full papers on every student start or image request (**A07**).
- Produce signed versioned manifests tying content, key, adapter, response schema, rendering, locale and decision lineage. Keep source numbers separate from student slots. Freeze canonical/display option mappings; do not shuffle source-image options unless the approved rendering supports it safely.
- Deliver usable staff review/approval/publish/withdraw/correct/revoke/regrade workflows, publish blockers and source-authoritative diagnostics. Catalogue/search indexes are replaceable discovery projections, never grading truth.

Interfaces/artifacts: source/content/version APIs, editorial ETags, reviewer/publisher decisions, immutable manifests, safe asset builder, rights/audience and solution-release policies, approved source corpus.

Exit gate: independent A02/A03/A04 regressions and mixed-paper/pairing/LaTeX/diagram/locale/rights/security tests pass; concurrent operations cannot lose work or duplicate versions. The original75-question paper is not published until actual academic, accessibility and rights approvals are supplied.

Packet 06 — Exam Engines, Durable Sessions & Exact Scoring

Depends on:01–05. **Owners:** session/scoring engineers and academic adapter owner.

Build/deliver

- Implement year/source-pinned **JEE Main, BITSAT, WBJEE and COMEDK** adapters from verified official material: sections, navigation, question counts/types, timers/accommodations, numeric syntax, multiselect/partial rules, bonus/invalid-item policy and BITSAT irreversible bonus transitions. Unsupported years stay unavailable; custom practice remains clearly separate.
- Atomically consume a signed owner-bound reservation, create one attempt/receipt and charge capacity. Freeze home cell, package, adapter, key, locale/display mappings and timing policy. Release capacity exactly once on terminal transitions; unconsumed reservation expiry cannot release an active slot.
- Close **A01** across ordinary starts, resumes and mistake retries: every successful operation key binds its complete canonical intent and durable outcome, including later replay after closure. Changed bodies/targets conflict; old receipts never overwrite newer accepted state.
- Implement explicit draft versus committed response semantics, all-or-nothing commit batches, single-writer generations, takeover, sequence/response-version rules and idempotent clears/review/phase changes. Return “saved” only after the engine's durable commitment—not cache or broker acceptance.
- At the attempt's fenced authority, acquire serialization lock then sample fresh authoritative time. Persist the complete admitted command and receipt; enforce the cutoff, seal the admitted watermark, drain/classify eligible work, and grade only the final committed snapshot. HTTP arrival time or local device timestamps do not establish deadline eligibility.
- Produce immutable exact-rational item/attempt ledgers asynchronously, with one score effect per attempt/key version. Preserve old grades, explicit approved correction versions and incident exceptions. Support reads/reporting must distinguish RECEIVED, APPLIED, SEALED, GRADING and READY.

Interfaces/artifacts: admission/start/lease/draft/commit/resume/submit/status/result contracts; command/receipt schemas; adapter registry; independent golden corpus; incident exception and regrade policy.

Exit gate: lost replies, committed-A/draft-B, clears, duplicate/no-change commands, stale writers, lock waits crossing cutoff, simultaneous submit/expiry, bonus transitions and reservation release all pass real-function/integration tests. Grade ledgers exactly reconcile. Cell-isolation/capacity qualification remains10.

Packet 07 — Frontend, Offline Recovery & Edge Experience

Depends on:01–06. **Owners:** frontend/accessibility engineers; CDN/edge/privacy operators.

Build/deliver

- Complete the matte-maroon landing, catalogue, phone entry, practice home, instructions/preflight, exam workspace, history, staff console, support/privacy and error/empty/loading flows. Use consistent design tokens/Shadcn patterns, unique test IDs, keyboard/screen-reader semantics, safe focus and44px mobile targets; fix template branding.
- Close **F03** with one safe math renderer for typed stems/options/solutions and accessible equivalents. Validate long integers, locale changes, shared passages and image-heavy questions on supported devices; never trade content fidelity for speed.
- Build a versioned offline capsule using suitable durable browser storage: candidate-safe assets, pinned schema/mappings, local drafts and reconciliation metadata, never answer keys/unreleased solutions. Verify/quarantine malformed/null/old records (**F01**) and explicitly scope discard/logout actions with pending-loss confirmation (**F02**).
- Enforce multi-tab leadership/writer-generation UX, precise pending/saved/conflict/expired states, bounded buffers and explicit recovery after refresh, backgrounding, auth refresh, slow responses and lost acknowledgement. Timer truth remains server-owned and offline time does not pause it.
- Prefetch verified immutable candidate assets only when question-release policy permits. Scheduled exams may prefetch encrypted candidate packages with capacity-tested time-gated access. Renew asset/session authority without synchronized expiry cliffs; never extend the attempt clock merely because a link expires.
- Apply route splitting, current-question virtualization, bounded caches and main-thread budgets. Jitter/coalesce retries and status polling; count all304s, renewals and reads as workload. Capture versioned visibility/confidence/revision/review events needed for08 without making optional telemetry a write dependency.
- Deliver source-authoritative staff/support diagnostics, bookmarks/notes and accessible review interactions through their proper normal/queued lanes; never allow a stale index to authorize or change grades.

Interfaces/artifacts: student/staff route map, component/state-machine contracts, storage migration/recovery protocol, edge authorization and caching rules, client event schema, browser performance profiles.

Exit gate: actual desktop/mobile/tablet/keyboard/slow-network tests close F01/F02/F03 or keep them explicitly blocked; no silent draft loss or false “saved”; candidate/key separation and scoped cached access hold. Additional client budgets are in the SLO document and require approval.

Packet 08 — All 60 Analyses, Cohorts & Study Plans

Depends on:01–07. **Owners:** analytics/data/frontend engineers; academic/privacy owners.

Build/deliver

- Implement **A01–A60**, not40cards: exact definition/version, inputs, denominator, evidence thresholds, output, interpretation, supporting-question drill-through and actionable next step for each metric. Close **A08**: unavailable evidence/zero denominators yield null with reasons, not invented zeroes.
- Build versioned visibility reconciliation, confidence/calibration, answer-revision impact, exposure, taxonomy, timing, longitudinal comparability, confirmed error causes and review histories. Observed visible time cannot exceed valid server elapsed time; missing observations remain explicitly missing.
- Stream authoritative grade/outbox events to isolated analytics stores; use idempotent contributions/retractions, preaggregated exact score histograms and materialized snapshots. No heavy cohort scans or joins on live attempt writers.
- Pin cohort eligibility to paper/item/key/adapter/mode/accommodation/exposure/time-window rules. Apply distinct-profile deduplication, minimum sample/privacy rules and complementary suppression. Use the guide's applicable thresholds, including privacy groups≥10 and percentile evidence≥100 where specified; distinguish privacy adequacy from statistical sufficiency.
- Preserve ties and fractional marks; never label platform percentiles as official ranks. Unverified phone profiles are not verified people; disclose cohort assurance and anti-abuse limitations.
- Deliver all results insight views plus reproducible7-day/30-day plans, prerequisite coverage, fresh review checks, notes/bookmarks/error notebook integration and honest evidence-specific unavailable states.

Interfaces/artifacts:60-entry feature/metric registry, versioned event/data contracts, golden datasets, cohort eligibility/suppression registry, materialized read APIs, study-plan/review APIs, regrade/replay reconciliation.

Exit gate: 60/60 positive applicable golden cases and missing/invalid-evidence cases pass; every ID has an implemented calculation, not a permanently unavailable placeholder. Duplicate/reordered/replayed/regraded events do not double-count; privacy and snapshot consistency pass. Full500k analytics isolation is qualified in10.

Packet 09 — Results, Released Solutions & PDF Reports

Depends on:01–08. **Owners:** results/report/frontend engineers; publication/privacy operators.

Build/deliver

- Bind UI, question review, subject/topic/time/cohort insights, correction comparisons, study actions and exports to the same immutable grade/content/metric/cohort snapshot identities. Paginate detailed review; never recompute from current editable keys.
- Release solutions only when both attempt policy and the signed release/rights policy allow it. In scheduled/shared papers, submission alone must not bypass embargo. An interim receipt/score view is not falsely labelled a complete solutions report.
- Build full source-faithful PDFs including every attempted/eligible question in the student's actual order, their response, key/marks and exactly matched approved solution, with version/provenance/incident metadata and defined accessibility/readability standards. UI filters do not silently remove questions from a complete export.
- Reuse reviewed rendering/templates, cache immutable snapshots, single-flight duplicate jobs and publish fenced immutable report pointers. Preserve original report bytes after correction; generate clearly labelled successor exports.
- Provide authorized version-specific downloads through scalable delivery rather than loading every large PDF through API memory. Test expired/revoked tokens, range/retry behavior, cross-profile isolation and cache authorization. Protect solution-consultation/exposure telemetry feedingA56–A60.
- Treat **all500k students requesting reports** as a workload scenario, not an assumed10%subset. Separate API receipt latency, grade readiness, PDF generation wait and download bandwidth. Show honest progress/retry states while isolated export workers run.

Interfaces/artifacts: immutable result/report snapshot contract, released-solution APIs, queued report/status/download APIs, safe delivery manifests, source-render parity corpus, export throughput/cost worksheet.

Exit gate: complete question–solution pairing, screen/export parity, embargo/access/grade-version tests and real75-question/full-length/large-render PDF cases pass. Proposed export budgets require measured capacity; missed PDF SLOs cannot be hidden behind fast job-creation responses.

Packet 10 — Full-Scale Qualification, Recovery & Operations

Depends on:01–09. **Owners:** QA/performance engineers, SRE/DBA/security operators, academic/product signatories.

Build/deliver

- Freeze the actual deployment/image/dependency/schema/package/adapter versions, resources, quotas, cost approvals, workload and failure scopes. Use dedicated nonproduction test capacity; never unleash500k traffic on this shared preview.
- Execute the guide's exact six stages: **1k/15min →10k/30min →50k/45min →100k/60min →250k/90min →500k/180min**. Warmup/admission is outside the final180-minute proof. Qualify the individual25k-cell envelope additionally; do not substitute it for these stages.
- Use independently reconciled stateful HTTP actors, a declared real-browser subset (guide example500–2,000) and separate observed-topology connection/churn tests. Arrival-rate demand must not fall when the service slows; fail saturated generators, dropped scheduled traffic and averaged localp99s.
- Test baseline,2×normal mix, pathological drafts, starts/submissions overlapping baseline, cold/warm assets, renewal/reconnect storms, status polling, all-report demand, hot-cell skew and platform quotas. Prewarm known waves; test30%API host loss and AZ/database/cell/control/queue outages separately.
- Verify exact accepted payload/sequence reconciliation, grades, destination fences, source/key safety, privacy and snapshot parity. Test alerts/collector failure with bounded labels; optional telemetry loss cannot masquerade as healthy monitoring.
- Run database/AZ failover, corruption and regional restoration at representative volume, including usable keys, manifests, receipts/outboxes, routing and writer fencing. Record actualRPO/RTO and affected windows; never claim zero regional loss for asynchronous replication.
- Complete signed IaC/image/SBOM/release artifacts and backward-compatible canary/rollback rehearsal with operators. Keep active pinned attempts compatible; drain before removing old schemas/leases. Publish cost/admission ledgers, support/incident runbooks and an operator-owned readiness decision.
- Retain **WP01–WP13 in order** and every **OPS-R01–OPS-R12** gate as internal checkpoints; these do not add top-level packets. See `QUALIFICATION_AND_HANDOFF.md`.

Interfaces/artifacts: signed run dossier, raw histograms/actor reconciliation, browser/connection evidence, fault/restore reports, quota/cost exports, release manifest, blocker register, incident/support playbooks, operator readiness APIs.

Exit gate: all mandatory gates PASS for the exact declared deployment and workload, including500kdistinct active attempts for180minutes, SLOs and covered-failure integrity. Any FAIL or UNQUALIFIED blocks the capacity claim. Only then may an authorized operator approve a staged production launch.

Cross-packet closure rules

- API, data, security, privacy, performance, migration/rollback and observability checklists apply to **every** packet.
- Each packet produces versioned design/contracts, implementation artifacts, exact tests, independent verification and known limitations. No source change inherits an old qualification automatically.
- Full requirement completion requires both its owning packet and its qualification packet to pass. A provisioned DB, a queue demo or60empty cards is not compliance.

- Coverage is in `coverage_manifest.json` and `COVERAGE_MATRIX.md`:62audit requirements, every named finding/risk/erratum,60analyses,177source pages,13WP checkpoints and12release gates. Packet01 expands page-level ownership into atomic clauses before implementation.
- No timeline, node count or cloud bill is invented here. Operator quotes, representative benchmarks and decisions determine capacity and schedule. These10packets are unequal gated milestones, not a promise that10short coding steps produce500k readiness.

Appendix A — Workload and SLO contract

Status: proposed acceptance contract, not measured performance. Packet01 freezes definitions and operator signoff; Packet10 produces qualification. Source guide numbers below are explicitly proposed by PDF-05, not vendor guarantees. Additional client/PDF budgets are separately labelled proposals.

1. Workload identities and boundaries

C=500,000 distinct admitted active attempts, T=10,800 steady-state seconds. Each has an independent actor/session, home cell, lease, adapter/package version, deterministic answer-state machine and expected receipt ledger. Idle users, cached duplicate operations and one identity reused500k times do not count.

Stream	Formula	Global baseline
Explicit commits	C/30s	16,666.7 requests/s
Dirty drafts	C×0.20/10s	10,000 requests/s
Lease renewal	C/60s	8,333.3 requests/s
Live subtotal	Sum above	**35,000 requests/s**
Optional telemetry batches	C/30s, isolated collector	16,666.7 requests/s
Combined baseline	Live+telemetry	**51,666.7 requests/s**

In three hours:180million commits +108million drafts +90million lease renewals =378million live operations; telemetry adds180million batches. These are not378million distinct answers, nor a direct IOPS estimate. Measure transaction/row/index/outbox/WAL/replication amplification.

A25,000-attempt cell carries1,750 live requests/s plus833.3 telemetry batches/s. Qualify3,500 live requests/s at the2×mix without increasing its authorized concurrency to50,000.20full cells provide no empty spare cell.

2. Separate overlays and capacity envelopes

Scenario	Required demand / measurement
2×normal throughput	70,000 live +33,333.3 telemetry =103,333.3 dynamic requests/s, **before** other overlays
Pathological draft case	100%dirty every5s →100,000drafts +16,666.7commits +8,333.3leases =125,000live/s; with baseline telemetry141,666.7/s. Separate test, not covered by2×normal
Start wave	500,000/120s =4,166.7starts/s; authentication, grants, preflight and content requests added explicitly
Submit wave	500,000/30s =16,666.7submissions/s; overlapping full baseline gives51,666.7live/s and68,333.3with telemetry, before retries/assets/other reads
Grade burst example	500,000×75/120s =312,500item evaluations/s. Recalculate for actual adapter mix, partial rules and expensive cases
Candidate delivery example	2MB×500,000 =1TB; over120s ≈8.333GB/s or66.667Gbit/s before overhead.2MB is a reference, not an assumption that image-heavy papers fit it
Report demand	All500k may request a complete report after grading/release. If within30s,16,666.7report requests/s **in addition** to other streams. Generation throughput, pages/s, bytes/s and storage/egress must be measured
Result/job polling	500k clients every10s would add50,000reads/s; every30s adds16,666.7reads/s.304responses still cost requests. Use bounded jitter/coalescing/materialized reads and include actual polling policy in the manifest
Renewals/reconnects	Independently measure retry amplification, expired asset/auth tokens, synchronized renewal cliffs, multi-tab churn and return-from-background traffic
Mixed traffic	Catalogue/search, staff operations, report reads, annotations, backups, telemetry and hostile/invalid traffic receive explicit quotas and measured overlays

Do not double-count or omit headroom. Declare exactly which overlays are simultaneous and whether their rates already include retry factors. Content cache-hit ratios reduce origin demand, not viewer bandwidth or authorization obligations. Cold and warm campaigns are separate.

3. Guide-backed proposed acceptance targets

Source: PDF-05 p55,61–62. All are **targets awaiting qualification**.

SLI	Proposed gate	Measurement boundary
Valid commit acknowledgement	p95≤300ms; p99≤1,000ms	In-region server ingress through queueing/validation/lock/DB durable commitment to acknowledgement. Not just handler CPU time or queue insertion
Durable submission receipt	p99≤2s under declared submit burst	Ingress to durable terminal receipt/seal-state acceptance; not PDF completion
Correct core grade readiness	p95≤60s; p99≤120s	Durable submission receipt to correct frozen core grade available. Includes grade queue wait and execution

SLI	Proposed gate	Measurement boundary
Server errors	≤0.1%overall **and in each predeclared critical window**	Distinguish infrastructure/server errors, expected business rejections, admission denials and cancellations. Overload failures of valid admitted operations cannot be hidden as harmless429s
Integrity	Zero violations within covered failure envelope	No wrong grades, duplicate score effects, lost acknowledged commits, wrong version/canonical mapping or forbidden source/key release
Covered AZ loss RPO	0, only with verified synchronous durable persistence	Reconcile acknowledged receipts against recovered durable truth. No standalone/asynchronous shortcut
Regional disaster	RPO≤5min; RTO≤60min	Guide proposal; declare nonzero async-loss risk. RTO includes detection, exclusive writer fencing, restore, key/object validation, routing and authorized essential-API usability

Freeze per-cell as well as global evaluation, the whole run, named start/submit/expiry/renewal windows and rolling windows (initial proposal5minutes). Never remove a failed window after observing results. Merge compatible latency histograms; do not average p99values across generators/cells. Publish tail latency and actor journey failures, not only HTTP success counts.

4. Additional proposed client/export budgets

These are planning proposals, **not numbers claimed from the PDFs**. Product/operator approval and representative measurements are required before they become exit criteria; if infeasible, make an explicit scope/SLO decision instead of claiming success.

Experience	Initial proposed budget	Conditions / guardrails
Browser responsiveness	INP p75≤200ms; CLS≤0.1	Agreed low/mid-range mobile and desktop profiles; real accessible mathematics and large-number inputs
Initial route experience	LCP p75≤2.5s on the agreed normal-mobile profile	Measure cold and warm separately; publish precise bandwidth/RTT/loss/CPU/cache definitions, not a vague “4G” label
Prefetched local question navigation	p95≤150ms; p99≤300ms	Only navigation not waiting for a required durable transition; never relabel server-dependent Save&Next as local-only
Student-observed save feedback	p95≤1.5s; p99≤3s on agreed normal connected profiles	Actual click→visible durable acknowledgement including network. Slow/offline profiles get truthful pending/recovery states, not a false saved badge
Complete PDF readiness	p95≤5min; p99≤15min from accepted export request	Includes queue wait/render/publish for the declared full paper mix and **all500k request** scenario. Grant/embargo may delay when export can lawfully be requested; do not start this clock only when a worker picks it up
Export/read traffic impact	Critical commit/receipt SLOs still pass	Grade, analytics and PDF workers/pools are separate; optional exports never steal answer-write resources

A75-question example with100pages/report and500kexports produces50million pages. Finishing all within900s would require≈55,556pages/s; this is an illustrative sizing calculation, not the mathematical consequence of a p99target or a measured renderer capability. Benchmark reusable templates,

personalization, storage and delivery before buying capacity. Publish actual achieved service, costs and shortfalls.

Proposed status polling must use server-directed backoff/jitter and a measured read budget; a minimum steady interval around30s is a starting proposal, not a hidden requirement that students see instant grades. Include initial result reads, manual checks and ready-to-visible delay explicitly. Optional persistent push channels require their own connection/authorization/failure proof; they are not free capacity.

5. Resource budgets and overload policy

Packet01/04 must freeze numeric per-schema/body/field/count/page/pixel/CPU/memory/concurrency/deadline budgets from a representative corpus. Initial upload guard proposals: one file/request,50MiBfile cap,52MiBtotal request cap,64KiBnon-file field cap and a small explicit field-count allowlist. Larger approved sources require a reviewed ingestion profile—not silently raised/unbounded limits. These are proposed limits, not a claim that current multipart enforcement works.

Reserve capacity by lane: **admitted commits/sealing first**, then lease/control essentials and core grading, then normal reads, then optional analytics/search/exports/telemetry. Drop/defer optional work explicitly; never drop a durable admitted command or display a cache write as saved. Missing telemetry becomes unavailable analytical evidence, not invented observations.

Admission depends on measured writer commit latency, pool wait, storage/WAL pressure, inflight work and pending registered commands—not API CPU alone. Stop new admissions before admitted attempts become unstable. Budget controls must never terminate admitted students to save money.

6. Node, database and failure sizing

For each service tier, with measured sustainable unit throughput μ and utilization factor u:

- Normal requirement: `n ≥ ceil(required_RPS / (μ × u))`.
- Separate30%equal-host-loss scenario: `floor(0.70 × n) × μ × u ≥ scenario_RPS`.
- Choose the maximum satisfying the declared scenarios; do not multiply blanket headroom twice. If μ was measured at the chosen utilization, use u=1.

Count physical host contention, database connection totals, storage/WAL/IOPS, standby capacity, object/KMS requests, network/NAT, LB/target limits, collector load and prewarming time. API-node survival is not DB-cell recovery. Twenty full cells do not magically absorb a lost cell's25kactive writers; restore/fence within that cell or use explicitly provisioned and tested migration capacity.

No machine counts or bill estimates are invented before benchmarks and dated provider quotes. Operators must reserve usable capacity/quotas ahead of the event; an autoscaling maximum or pending quota request is not readiness.

7. Exact qualification progression

Source: PDF-05 p58. Each stage needs reviewed evidence before promotion.

Active actors	Minimum steady-state duration
1,000	15minutes
10,000	30minutes
50,000	45minutes
100,000	60minutes

Active actors	Minimum steady-state duration
250,000	90minutes
500,000	180minutes

Additionally qualify a25kcell's baseline/2×mix and skew before fleet claims. This extra cell test does **not** replace the six stages. For final proof, all500k admitted attempts remain active throughout the180-minute steady period; warmup, preflight and admissions do not count. Use scheduled-start fixtures. Run submit/baseline overlap scenarios separately where necessary rather than terminating actors early and calling it a full180-minute active soak.

Three noninterchangeable tracks:

1. Stateful HTTP actors with independent expected response/receipt/grade ledgers.
2. Declared real-browser subset (guide example500–2,000) during saturation, with real devices/network profiles and accessibility interactions.
3. Measured connection/socket/handshake/churn inventory based on origin multiplexing.500kactors do not prove500kTLSsockets.

Arrival-rate generators must sustain offered load; report scheduled/started/completed/dropped operations and scheduler delay. An under-generated, interrupted or generator-saturated run FAILS. Cold, warm and fault scenarios retain separate results. Fault envelopes include lost replies, duplicate/out-of-order commands, expiry cliffs, stale writers, poison jobs, control-plane isolation, host/AZ loss, DBstall/quorum loss, cache/object/KMS failures, regional restore and telemetry overload.

8. Versioned run manifest and release decision

Record: source/image/SBOM/schema/config digests; exact DB/service versions; cell/resource/pool layout; actor ranges/seeds/exam mix; payload/asset sizes and cache states; intervals/dirty fractions/retry factors; official scheduling/cutoffs; browser/network profiles; connection inventory; SLO definitions/windows; fault scopes; evidence locations; cost/quotas; operators and approvals.

The final dossier includes per-actor reconciliation, histograms, pending/backlog age, WAL/replication/storage metrics, browser states, current key/package authorization, restore timelines and cost. No wrong grade or covered lost acknowledgement may be waived because latency looks good. Material code/schema/config/workload changes require requalification.

Nothing in this document states that these targets have been achieved.

Appendix B — Qualification and operator handoff

This is an appendix to the **exactly10top-level packets** in `TEN_PACKET_STRATEGY.md`, not additional packets. No infrastructure, cloud purchase, external message, credential rotation, deployment or application change is authorized by this strategy document alone.

1. Roles and responsibility split

Owner	Responsibility
Product owner	Scope/SLO/failure-envelope approval, student phone-only trade-offs, launch decision and disclosed limitations
Academic owner + independent reviewers	Official-year evidence, source rights, exact pairing, taxonomy, accessible content, anomaly/key policy and independent publish/correction approval
Application engineers	Domain boundaries, typed APIs, repository interfaces/migrations, session/scoring/content/UI/analytics/report logic, instrumentation and isolated regression tests
QA/performance engineers	Independent expected ledgers, real-function regressions, honest mock labels, generator/browser/connection evidence and complete acceptance coverage
Cloud/SRE/DBA operators	Actual DB/server/network/worker/CDN/WAF/KMS/backup provisioning; quotas/prewarming; IaC/CI/CD/SSL; production deployment, failover and restore operations
Security/privacy owner	Staff assurance, grants/key custody, audit/retention/legal policy, telemetry minimization, incident response and exposed-credential rotation

The builder must not quietly administer infrastructure, change existing environment connection values or perform production deployment. Missing operator capacity/credentials/evidence is an explicit dependency, not an application implementation success.

2. Decisions to freeze before their dependent packet

Decision	Required before	Required evidence / constraint
D01 scope, exact SLOs and covered failure modes	01exit	Guide-backed targets plus approved extra client/PDF targets; no silent weaker substitutes
D02 cloud/provider, region/data residency and accountable operators	02start	Current official support/limits, latency/legal needs, dated quotes, access responsibilities
D03 exact PostgreSQL HA/replication/failover variant and migration policy	02exit	Durable acknowledgement semantics, standby/read/failover behavior, writer fencing, rollback and legacy protection. No generic “Multi-AZ” promise
D04 staff identity/MFA, managed signing/secrets and PIN policy	03start	Provider selection, secure credential channels and lifecycle. No credentials in reports or code
D05 academic authority, official years, paper rights, independent reviewers and release/embargo rules	05/06start	Verified current primary sources and actual human approvals; unsupported years unavailable
D06 resource budget, quotas, per-cell specification and synthetic-test spend	02/10	Operator approval, usable quota reservations, provider limits, measured unit curves—not guessed node counts
D07 device/network/browser and export-demand profiles	01exit,07/09freeze	Exact bandwidth/RTT/loss/CPU/cache definitions; full500kexport scenario acknowledged
D08 privacy/consent, retention, source PIIdetection and product telemetry	03exit	Legal/operator-approved matrix; privacy-managed learner records separate from locked academic artifacts; review PostHog configuration with platform owner
D09 writer takeover, account recovery and irreversible deletion under phone-only access	03/06exit	No implicit verified-person promise; student login stays phone-only; destructive support actions need an approved assurance policy
D10 outage fairness/remedy, regional loss disclosure and incident authority	06/10	Auditable approved exceptions; no silent deadline rewrite or invented restored answers
D11 credential containment	01	Authorized persisted-account rotation and relevant session revocation; the audit's redaction is not credential rotation

Service/model/library versions must be verified against current provider guidance before integration, with proper playbooks and credentials collected securely. No provider keys are requested or used merely to write this strategy.

3. Preserve all13operational work packages inside the10packets

PDF-05 p65–66 names13**sequential operational gates**. Build prerequisites in the owning packets, then retain formal WP01→WP13 verification/signoff order in the qualification dossier. This preserves the guide without creating13top-level implementation packets. Release artifacts must be generated/pinned before tests; WP12 verifies the **same tested artifacts**, not newly rebuilt untested images.

All example routes below are contract examples to map under `/api` (e.g. `/api/v1/...`, `/api/internal/...`) with explicit legacy aliases. Internal operations are operator-authenticated and not student-accessible.

Gate	Deliverable / check	Artifact builders	Formal verification
WP01	Workload/SLO manifest; internal capacity profile; arithmetic/time boundaries	01,02	10, before traffic qualification
WP02	Independent actors, adapters/options, expected response/grade fixtures; starts	01,05,06	10
WP03	Durable commit tests; lost ack, clears, duplicates; accepted-payload reconciliation	02,06	10
WP04	Distributed generators/run status; actor partitions/offered load/scheduling capacity	01,10	10
WP05	Start/submit/cold-cache/expiry overlap and recoverable receipts	06,07,09,10	10
WP06	Nonproduction fault suite; AZ/DB/queue/control outages and receipt reconciliation	02,03,04,06,10	10
WP07	Dashboards/collector health/alarms; bounded cardinality and collection overhead	01–09	10
WP08	Restore/failover; usableKMS/packages, exclusivewriterfence, rational rescore, actualRPO/RTO	02,03,06,09,10	10
WP09	Cost/admission ledger; quota/budget controls protect admitted students	01,02,04,06,10	10
WP10	A01–A60registry, versioned reports, unavailable states, embargoes/cohort revisions, no hot analytical scans	05,07,08,09	10
WP11	Staff/support source-authoritative diagnostics and cell-aware receipt APIs; stale-index tests	03,05,06,07,09	10
WP12	Operator-reviewedIaC; signed images by digest/SBOM; compatible migrations/rollback	01–09 plus operators	10, matching already-tested digest/config
WP13	Six-stage dossier/blocker register;500k/180minute evidence and named operator signoff	All10packets	10, last

4. Keep all12production-blocking gates

Source: PDF-05 p67–68. Every gate has a named operator, artifact references and retest triggers; **FAIL or UNQUALIFIED blocks the capacity/release claim**.

Gate	Pass condition
OPS-R01	500,000distinct active actor ledgers for180steady minutes, not idle users
OPS-R02	Intended offered demand; unsaturated generators; correct public-edge path; merged histograms; no hidden dropped traffic
OPS-R03	Zero wrong grades, duplicate score effects or committed-A/draft-B violations; every accepted payload reconciled
OPS-R04	Covered crash/AZ failures preserve acknowledged truth beyond volatile caches/queues

Gate	Pass condition
OPS-R05	Start, submit, cold-cache, expiry and reconnect windows meet declared gates, not just a global average
OPS-R06	Exclusive writer fences, usable keys and validated restored version truth; no split brain
OPS-R07	Measured regionalRPO/RTO with explicit asynchronous-loss scope and reconciliation before grade release
OPS-R08	Report/cohort/grade snapshot consistency, embargo enforcement and no duplicate contributions
OPS-R09	Bounded metric labels, complete durable receipt evidence, working alerts and collector-health truth
OPS-R10	Qualified reservations, approved quotas/test costs and budget controls that do not terminate admitted students
OPS-R11	Signed digests/SBOM/reviewedIaC and compatible rollout/rollback; no ungated production source changes
OPS-R12	Auditable incident exceptions and source-authoritative support; no silent time rewrites or stale-index grading

5. Audit findings and permanent closure owners

All remain **OPEN** until implementation and independent verification. Packet01 captures the exact regressions and applies only explicitly authorized containment.

Finding	Primary repair packet	Required proof
A01 retry-resume key reuse	06	K1→R1;K2resumesR1;closeR1;K2replayreturnsR1;changedintentconflicts
A02 same-version publication race	05	Concurrentpublish yields one transition/archive identity; replay outcome stable
A03 reviewer lost update	05	Staleexpectedrevisionconflicts and preserves first edit, including different-question edits
A04 audience fail-open; A04b expiry validation	05	Unknown/invalid/audience/date failsclosed on creation/update/read; validrestricted behavior preserved
A05 idle queue; A05b recovery fairness	04	Queued work drains without polling; poisoned/failed head entries cannot starve later work
A06 stale extraction writer	04, integrated05	Leasechangeaftercheck/beforewrite cannot publish; destinationauthorityenforced
S01 multipart/resource gap	04	Actualmultipart(nonfile/file/count/total)limits enforce before expensive work; legitimateuploadswork
F01 malformed local state (unverified)	07	ActualPlayerbrowsernull/primitives/malformedrecordrecovery; not just a language expression
F02 logout draft scope	07	Explicit current-profile discard; no silent other-profile loss; pending warning/recovery
F03 typed option math	07	Safe correct math/accessibility on all supported viewports and hostilemarkupfixtures

Finding	Primary repair packet	Required proof
A07 live-path parsing/packing work	04/05	Publishedassets/packagesreused; no sourceparse/repack on live start/read path; measured event-loop budget
A08 zero-denominator metrics	08	Null+typedunavailable reason where defined; positive/missing evidence goldens
Exposed audit credential	01 operator	Actual persisted credential/sessionrotation proof; no secret values in output
Dependency advisories	01 triage;04/07 closure	Current advisories/SBOM, exploitability/risk decision, compatible fixes and regression tests; no unreviewed critical exploitable path at release
Undisclosed/unreviewed instrumentation	03 policy +07 integration	Approved privacy/consent/minimization and platform-supported configuration; protectedcontent excluded
Template title, touch targets, inconsistent controls/testIDs	07	Approved branding,44px targets, keyboard/screen-reader checks, stable unique IDs
Harness isolation and false pass claims	01; enforcedeverypacket	Real implementation calls, fail-closed isolated repositories, independentoracles, scopedproof, credential-safeartifacts

6. Evidence package required for every packet

- Approved source clauses and explicit accepted overrides; no orphan requirements or hidden deferred fixes
- Input/output, permission, data-owner, versioning, concurrency and idempotency contracts
- Implementation and dependency digests, migration/backward-compatibility and rollback plan
- Golden inputs, independent expected outcomes, negative tests and real-function/browser/integration evidence; clear **MOCKED TEST DATA** labels where appropriate
- Original failing audit regression plus **testing-agent** verification for every repaired issue
- Performance/resource/cost evidence appropriate to the packet; no scope inflation into500k claims
- Privacy/security/dependency checks, observability and operator runbook; no raw credentials or unnecessary student data
- PASS/FAIL/UNQUALIFIED status, unresolved blockers, owner and explicit promotion decision

Infrastructure, cloud accounts, quota approvals, cost funding, identity/KMS selections, independent academic reviewers and a representative corpus cannot be replaced by code or a screenshot. If any prerequisite is unavailable, mark the corresponding gate blocked rather than calling the packet fully delivered.

Appendix C — Full architecture audit

Verdict: DO NOT SIGN OFF for a student launch or 500,000-concurrent-user operation.

This is an **audit, not a repair release**. The project has useful working practice components, but the audit found reproducible correctness defects, incomplete requirements, and unqualified infrastructure. Earlier “hardened”, “fenced”, or high test-pass claims must not be read as full compliance.

1. Scope, authority, and preservation

User instruction: audit the architecture and everything coded against the updated rules **before changing application code**.

- Authoritative source: all five supplied PDFs, **177 pages**; original URLs and SHA256 values remain in `../source_guides/manifest.json`.
- User overrides retained: **matte maroon**, and **phone-number-only student access without OTP/password**. These are deliberate product choices, not findings to “fix”. Staff access remains separately assessed.
- Initial source baseline: **97 application/configuration files, 29,382 lines**, in `source_baseline.json`. Four additional entry/build files were subsequently inventoried in `supplemental_source_inventory.json`.
- Reviewed custom backend modules, custom React pages/components/hooks, source/grade/report flows, configuration, dependencies, tests and existing claims. Shadcn primitives were reviewed at integration points, not formally verified line by line. An inventory is not a code-coverage measurement.
- No application-source, connection environment, infrastructure, credentials, published paper, answer, or report was repaired during this audit. Original-source SHA256 comparisons are recorded separately in `source_integrity_final.json`.
- Audit-created tests and documents are new artifacts. One rejected harness accidentally created four unreferenced test-only content packs outside its namespace; those exact records were identified and removed. See `harness_side_effect_inventory.json` and `harness_cleanup.json`. Subsequent corrected tests show unchanged main-collection counts and no remaining audit collections.

2. Evidence policy

Evidence class	Meaning
R	Actual application function executed against isolated Mongo collections; dependencies outside the tested behavior mocked and disclosed
B	Actual built frontend exercised in a browser; controlled synthetic attempt responses are **MOCKED TEST DATA**, not a real student's attempt
H	Real application middleware with installed framework exercised in a local ASGI harness; not a public load or exploit test
S	Source/configuration inspection with file and line references
D	Dependency advisory scan; establishes a version match, not exploitability
N	Not implemented, not provisioned, or not verified; these are distinct from a reproduced runtime bug

No overall “architecture percentage” or “500k readiness score” is assigned. A test that reproduces a bug has succeeded as an audit test; it does **not** mean the application passed its requirement.

Reports 24 and parts of 25 were rejected because tests copied logic, missed the intended sequence, or failed to patch the real collection object. Reports 27–29 also contain overclaims corrected in `EVIDENCE_LEDGER.md`. The validated records below take precedence over generic “100% passed” sections in subagent output.

3. Actual architecture, not the proposed one

```
Browser React SPA
  -> one FastAPI application / API boundary
      -> shared MongoDB writer and shared identity/content/attempt stores
      -> in-process grade/report/extract/regrade execution limits
      -> existing private object-storage gateway
      -> local source/pack/crop caches
Platform cron -> periodic repair/dispatch of a bounded batch
```

Mongo's read-only `hello` response confirms **no replica-set name and no mongos** in this environment. A requested majority/journal write concern on standalone Mongo is not multi-AZ acknowledged-write durability.

The code is a **modular monolith**, not 20 independent live cells. File boundaries are useful, but they are not independent database writers, queue failure domains, workload identities, admission budgets, or outage boundaries.

PDF-05's 20 × 25,000-cell example, 35,000 live RPS baseline, 2× envelope, and separate pathological burst model are **planning inputs only**. There is no representative single-cell saturation result, distributed 500k run, AZ failover, quorum-loss test, or restore drill in this audit.

4. Reproduced application findings

Severity: **P1 = blocks release of the affected correctness/security capability**; **P2 = significant functional/UX correctness gap**. These are not CVSS scores. All findings remain **OPEN; no fixes applied**.

A01

Severity: P1

Finding and observed result: Retry-resume identifiers are not durable. K1 creates retry R1; K2 resumes R1; after R1 completes, replaying the already-successful K2 creates a different R2. Ordinary start aliases do not solve this separate retry path.

Code / guide: `backend/tejas/attempts.py:388–430,471–490`; V5 p29

Evidence: R: report 26, real `retry()`

A02

Severity: P1

Finding and observed result: Concurrent publication duplicates one version. Two real publication calls succeed and create two archives for the same set/version. The revision guard is not advanced by publication, and no unique set/version archive constraint prevents this.

Code / guide: `content.py:565–633`; `db.py` indexes; V3 p20–21, V5 p17–20,29

Evidence: R: report 26, real `publish()`

A03

Severity: P1

Finding and observed result: Reviewer edits silently lose updates. Two real edits succeed; the later update overwrites the first. There is no client expected-revision condition on this edit mutation.

Code / guide: `content.py:472–521`; V3 p13–14,20; V5 p28–29

Evidence: R: report 26, real `edit_question()`

A04

Severity: P1

Finding and observed result: Unknown audience values fail open. Real `create_set()` accepts an invalid audience. `permitted()` then treats it as non-restricted, and the real detail handler allows access to a record made published inside the isolated fixture. Creation does not whitelist the enum even though updates validate it. This is not evidence that an existing private paper leaked publicly.

Code / guide: `content.py:40–53,84–98,159–172`; `attempts.py:114–119`; V3 p2–3; V5 p11–13

Evidence: R: report 26, real create/policy/detail handlers

A05

Severity: P1

Finding and observed result: A freed worker lane does not drain waiting work. With two grade jobs deliberately held running, a third dispatch remains queued. After both finish, it stays queued until another explicit dispatcher call. Result polling and the periodic cron can later recover it, but they are not a continuously serviced durable worker fleet.

Code / guide: `jobs.py:50–60,131–132,169–188`; `.emergent/crons.yml:2`; V5 p31,46,52

Evidence: R: report 26, real `run_job()` and event-held capacity

A06

Severity: P1

Finding and observed result: Extraction fencing has a check/write race. The real worker passes `require_lease`; its lease is replaced before its set update; the old worker still modifies the draft because the destination condition includes revision but not a fencing generation. A later job-completion check does not undo that content write.

Code / guide: `jobs.py:45–47,89–94`; V3 p14; V5 p10,40,50

Evidence: R: report 26, real worker and intercepted real collection write

S01

Severity: P1

Finding and observed result: The JSON size guard is not an upload/form budget. With the actual `JsonBodyLimits` middleware and installed Starlette parser, a 2,097,152-byte non-file multipart field is accepted by a local ASGI endpoint. The installed Starlette version also matches known multipart DoS advisories. A live outage/exploit was not attempted.

Code / guide: `http_limits.py:30`; `content.py:245–271`; `requirements.txt` Starlette 0.37.2; V5 p39–40,52

Evidence: H: corrected report 30; D: OSV

F02

Severity: P2

Finding and observed result: Sign-out erases other profiles' pending drafts. The logout action deletes every `tejas-draft-*` key, not just the current profile. Two scoped synthetic drafts were removed without an immediate loss confirmation. The privacy page does disclose that logout removes drafts; that disclosure does not resolve the cross-profile scope or interruption UX.

Code / guide: `frontend/src/lib/auth.js:11`; `features/Info.js:117–123`; V2 p17; V5 p41–42

Evidence: B: report 27; S: explicit key filter

F03

Severity: P1

Finding and observed result: Typed answer-option mathematics displays raw markup. A controlled Player fixture renders options such as x^2 literally; option content is not passed through the math renderer. This concerns typed options, not source-image options.

Code / guide: `features/Player.js:253–295`; `components/QuestionContent.js`; V3 p5–7,22; V5 p42–43

Evidence: B: report 28, both required viewports

Additional code-level risks not promoted to reproduced runtime findings

- **F01 — malformed/null local draft recovery:** `useAttempt.js:45–57` accepts parsed `null`, then calls `Object.entries` without validating a record shape. The language expression throws, but the actual Player browser observations were inconsistent. Status: **NOT VERIFIED at UI level**, not “fixed”. Quarantine/recovery behavior needs a correct browser regression.
- **A04b — malformed audience expiry:** creation accepts a free-form expiry string; comparisons use strings. Updates parse dates. This adjacent creation-validation path needs an isolated date-boundary test before a separate runtime finding is claimed.
- **A05b — repair backlog fairness:** maintenance inspects bounded, unsorted first batches and is not a fleet dispatcher. Failed/unserved entries can consume the recovery window; high-volume starvation has not been load-tested.
- **A07 — hot-path CPU/source work:** every new attempt still reconstructs/hashes/compresses and upserts a question pack through `packs.seal`; staff source-view handlers parse PDF data synchronously. Limits and local caches reduce some work but do not prove event-loop or CPU isolation. This is source evidence, not a measured latency regression.
- **A08 — analytics zero denominator:** `scoring.py:103–119` returns zero for unavailable accuracy/percent/rate denominators, while V4 p5–6 requires null/unavailable for the corresponding undefined metric. Marks correctness and metric semantics are different requirements.

5. Security and dependency review

Immediate audit-process incident — credential rotation required

A testing agent included an existing staff credential in report 27. The saved artifact was redacted; `credential_redaction.json` contains paths and actions, not the value. Prior tool-log output cannot be retracted.

Required authorized action: rotate the actual staff account credential and revoke relevant sessions. Merely changing a bootstrap environment value may not change an already-created account. **No credential or session revocation was performed during this audit.** This incident is separate from the application defects above; it is not proof of external account compromise.

Dependencies

- `yarn audit`: **51 advisory records, 27 unique module/advisory pairs**; reported record severities: 21 high, 25 moderate, 5 low. Many concern the CRA/webpack build or development toolchain, not necessarily deployed browser execution.
- OSV query checked **12 selected installed Python packages**, not the entire environment. Starlette 0.37.2 matches multipart-form resource advisories. File-spooling, range-response and unrelated dependency advisories were not independently exploited.
- KaTeX 0.16.22 matched an advisory requiring pre-existing prototype pollution to bypass trust restrictions. No such pollution/XSS chain was demonstrated in this application.
- Local HS256 configuration, package presence, or an advisory count alone does not prove reachability of JWK/algorithm-confusion vulnerabilities. Runtime reachability must be triaged before risk claims or upgrades are declared verified.

Third-party instrumentation and secrets boundaries

`frontend/public/index.html:64–133` initializes PostHog. The public browser showed `autocapture=true`, `capture_pageview=true`, `opt_out_capturing_by_default=false`, `disable_session_recording=false`. These flags establish configuration, **not proof that sensitive question content or credentials were exported**, nor proof that session recording is currently active server-side.

Protected study/staff surfaces need an explicit telemetry inventory, consent/minimization policy and production configuration review. “No advertising trackers” is not equivalent to “no product analytics”; the existing copy does not fully describe this instrumentation. Platform/editor scripts must be handled with the platform owner, not removed blindly during an audit.

Staff password/RBAC separation exists. Required SSO/MFA, payload-bound one-use step-up grants, independent approval, KMS/HSM signing, WORM/Object Lock, per-artifact-class workload identities and production key lifecycle are not implemented/provisioned.

6. Full requirement status

The detailed matrix is `REQUIREMENTS_MATRIX.md`; all 60 analytics entries are in `analytics_60_matrix.csv` and `.json`.

Area	Status	Interpretation
Exact marks for currently supported custom MCQ/integer practice	PASS, scoped	Seven real scorer fixtures passed, including wrong-answer penalties, signed integers, equivalent key, dropped/award-only items and exact fractional marks. This is not approval of every official exam adapter.
Acknowledged save receipts / ordinary start aliases	PARTIAL	Useful atomic properties exist; the retry-resume path fails A01; no complete distributed command register/lease protocol.
Content pairing, private source handling and attempt snapshots	PARTIAL	Reviewed refs, hashes and private API gates exist; physical release classes, signed manifests and four-eyes protected activation do not.
Publication and reviewer integrity	FAIL	A02/A03/A06 block a claim of safe concurrent editorial operation.
Batch processing	PARTIAL / FAIL at required guarantees	Heavy grade/report work is off the request path, but A05/A06 and process sharing prevent a reliable isolated worker-fleet claim.

Area	Status	Interpretation
Phone-only student access / matte maroon	PASS, user override	Preserve these decisions; do not reintroduce OTP or the old blue palette as an audit fix.
Offline practice UX	PARTIAL	Pending draft recovery and countdown have useful behavior; logout loss, malformed-record recovery and a full offline-open capsule remain gaps.
Formula fidelity	FAIL for typed options	F03; unrelated source-image rendering is not the failing path.
Results / PDF snapshots	PARTIAL	Core score and complete-report paths exist, including separate correction versions. Full report/accessibility/analytics specification is not complete.
All 60 analyses	FAIL	Ten related analysis contracts have partial implementations; fifty were not found as metric calculations plus metric-specific unavailable-state contracts. No full analysis PASS is asserted. This is not a percentage score.
API contract-first boundary	PARTIAL	`/api` envelopes exist; AST inventory found 46 routes but only two explicit response models, with ten untyped dictionary-body handlers. Versioned envelopes are not a complete typed API.
Database/cells/durability	FAIL against the target architecture	One standalone writer; no independent cells, transactional distributed admission, generation fencing or multi-AZ durability proof.
500k capacity, chaos, recovery and accessibility certification	NOT VERIFIED	No qualifying campaigns or certifications exist in this evidence set.

7. Rulebook errors and claim corrections

See `RULEBOOK_ERRATA.md`. Most importantly:

1. **V4 p1 mandates all 60 analyses**, and V5 p64 repeats this. The internal rulebook's “at least 40 meaningful analyses” relaxes the actual specification without a user-approved override.
2. V2/V5 require an explicit draft/commit/admission/seal protocol. The implemented custom autosave policy is labelled honestly, but labelling does not make it compliant with that protocol.
3. A lease read before a write is not a write fence (A06). Separate semaphores are not independent worker fleets (A05).
4. Private objects plus hashes are not independently signed academic truth, independent approvals, or WORM retention.
5. High pass percentages in small test reports are neither comprehensive coverage nor capacity qualification. Some prior “passes” were no-op, malformed-input, copied-logic or selector errors.

8. UI and operational observations

- Public desktop1920×800 and mobile390×844 screenshots had no horizontal overflow; settled matte-maroon dark-mode controls were readable.

- Browser title remains **“Emergent | Fullstack App”** (`public/index.html:24`). Two measured mobile targets were below44px high: brand link40px and login42px. These are P2 polish/accessibility gaps, not evidence that the whole UI is broken.
- Native controls and literal style values remain throughout the custom UI. Shadcn adoption/tokenization is partial; duplicate test IDs occur in repeated components. No complete keyboard/screen-reader/zoom-device audit was completed.
- Current catalogue: **50 QA sets withdrawn; one original75-question paper in review, version1/revision3**. No approved student paper is published. This is an intentional approval gate, not a fixture-access defect to bypass.
- Frontend production build and Python compilation pass. Build success does not invalidate the correctness findings.

9. Remediation order — proposed only

Immediate authorized security action

Rotate the exposed staff credential and revoke sessions; retain a redacted incident record. Review telemetry policy and prevent credentials from entering future test output.

Gate1: correctness before publication or scale claims

Fix A01–A06 and S01; then address typed-option math and safe draft-discard scope. Each fix needs the exact failing regression converted into a passing requirement test, plus a new **testing-agent** verification report. Do not weaken permissions or substitute copied logic to get a green result.

Gate2: academic truth and operator controls

Implement immutable publication identity/CAS/uniqueness, segregated approval, full typed contract families, staff step-up and signed release artifacts. Treat source parsing as hostile work in a physically isolated execution boundary.

Gate3: per-cell architecture and durability

Select/provision the required writer and durability topology, transactional admission/command authority, generation fences, terminal drain rules, autonomous batch dispatch and reserved capacity. Reconcile the Mongo pilot with the guide's relational cell protocol through an explicit design/migration decision—not a silent equivalence claim.

Gate4: complete product and evidence

Official year-pinned adapters, full60-analysis registry/calculations/unavailable states, all report/solution-release workflows, accessibility, source review. Then qualify one cell, test skew and failure, progress to distributed capacity and restore campaigns. Admit only a measured envelope.

No remediation has been applied. Findings and acceptance tests are ready for review before implementation.

Appendix C.1 — Rulebook errata

Source precedence: the five uploaded PDFs, except explicit latest user overrides (matte maroon and phone-only student entry). Internal documentation cannot silently lower a requirement.

E01

Current internal statement / omission: `docs/RULEBOOK_ARCHITECTURE_V5.md` §6 says at least40 meaningful analyses

Authoritative requirement: V4 p1: **Deliver all60 numbered analyses**, with informative unavailable states when evidence is missing. V5 p64 acceptance table: all60calculations match reference cases

Required correction: Replace the40minimum with all60; distinguish implementation/unavailable evidence rather than omit analyses. The audit extracts all60 entries in `analytics_60_matrix.csv`

E02

Current internal statement / omission: Custom autosave is an acknowledged pilot exception but sometimes discussed alongside full session correctness

Authoritative requirement: V2 p13–17 and V5 p7,10,29–31 require draft vs explicit commit, command identity, fenced authority, admission and terminal drain rules

Required correction: Keep custom-practice disclosure, but mark full official session protocol noncompliant until implemented; it is not a user-approved replacement for the guide

E03

Current internal statement / omission: Prior “lease-fenced jobs” language can imply protected outputs

Authoritative requirement: V5 p10,40,50: old writers must not publish after authority changes

Required correction: Require a destination-side generation/token condition or equivalent transactional fence. A06 proves check-then-write is insufficient

E04

Current internal statement / omission: Bounded lanes described as a scale enabler can imply independent queue execution

Authoritative requirement: V5 p44–46,52: separate failure/resource domains and functioning queue service

Required correction: State explicitly that current slots are per-process limits and queued work does not self-drain after capacity is freed (A05)

E05

Current internal statement / omission: Existing response envelope described as a versioned API boundary

Authoritative requirement: V5 p28: complete versioned OpenAPI/contracts, bounded schemas, errors, idempotency, status/header conventions

Required correction: Record46routes/two explicit response models as partial, not contract-first completion

E06

Current internal statement / omission: Immutable/hash snapshots can be mistaken for protected academic truth

Authoritative requirement: V5 p17–27: independent approvals, signed immutable activation/lineage, key custody and storage retention controls

Required correction: Separate content-addressed snapshots from cryptographic attestation, WORM and authorization boundaries

E07

Current internal statement / omission: Positive scorer fixtures / pending-grade tests treated as overall analytics proof

Authoritative requirement: V4 p5–6 requires null/unavailable for zero denominators, full outcome/award decomposition and evidence metadata

Required correction: Test metric semantics separately from net marks. Current zero-default accuracy/percentage behavior is not proof of correct unavailable states

E08

Current internal statement / omission: Legacy reports use broad success percentages or prose “all good” conclusions

Authoritative requirement: V5 p64–68 requires reproducible golden, load, fault, restore and gate evidence

Required correction: Preserve raw reports but bind conclusions to tested scopes; supersede copied/no-op/incorrect-encoding tests, never present their percentages as readiness

E09

Current internal statement / omission: Scope casually called “vaulted/modular/scalable”

Authoritative requirement: V5 p44–50 separates live cells, control/trust, batch, analytics, edge and operations

Required correction: Call the current deployment a modular monolith with a standalone writer. The reference topology is unprovisioned and unqualified

These are audit findings. The binding source PDFs, user decisions and application code were not edited during this audit. A corrected implementation/rulebook revision should be approved as a subsequent change set.

Appendix D — All 62 audited requirement rows

Scope: five supplied PDFs + explicit user overrides. This is not a release certificate. PASS means only the narrowly stated row is supported; PARTIAL means a useful subset exists; FAIL means required behavior is absent or contradicted; NOT VERIFIED means evidence is missing or the relevant infrastructure is outside visibility. See `AUDIT_REPORT.md` for reproduced findings and evidence classes.

Product, frontend and session behavior

R01

Requirement / source: Latest matte-maroon override, replacing V1 blue palette

Status: PASS, scoped

Current implementation / evidence: `App.css`, `index.css`, `ThemeToggle.js`; real light/dark screenshots at1920×800 and390×844. Not a full contrast certification

R02

Requirement / source: Latest phone-only student entry, no OTP/password

Status: PASS, user override

Current implementation / evidence: `phone.py`, `Auth.js`; explicit unverified-profile disclosure. Do not flag the deliberate lack of verified phone ownership as an accidental bypass

R03

Requirement / source: Separate student/staff authorization, V1 p10; V5 p9,13

Status: PARTIAL

Current implementation / evidence: Staff password/RBAC exists and phone profiles cannot elevate through staff flow. SSO/MFA and protected step-up absent

R04

Requirement / source: Full official-year adapters and versioned rule provenance, V1 p9; V2 p2,24–25

Status: FAIL

Current implementation / evidence: `profiles.py` pins custom profiles; only JEE MCQ/integer and BITSAT MCQ practice are implemented, not complete official-year engines

R05

Requirement / source: WBJEE/COMEDK and full BITSAT phase semantics, V2 adapter chapters

Status: FAIL

Current implementation / evidence: Not implemented; no multiselect/partial-credit adapter or bonus-phase irreversible transition. These are specification gaps, not broken implemented menu items

R06

Requirement / source: Explicit draft/commit and navigation semantics, V2 p13–14; V5 p29–31

Status: FAIL

Current implementation / evidence: `answers.py` treats acknowledged saves as scoring-eligible under custom autosave. No complete separate explicit commit register

R07

Requirement / source: Final accepted response is authoritative, V2 p18

Status: PASS, scoped

Current implementation / evidence: Seven actual scorer fixture baselines in report29; submitted local pending data is not substituted into frozen marks

R08

Requirement / source: Body-bound durable save retries, V5 p29

Status: PARTIAL

Current implementation / evidence: Answer receipts and ordinary start aliases exist; retry-resume fails A01. No complete mutation-wide operation catalogue

R09

Requirement / source: Monotonic sequencing / post-lock authoritative cutoff, V5 p10,29–31,50

Status: PARTIAL

Current implementation / evidence: Per-attempt response seq + single-document Mongo conditions exist; no relational lock-time authority/contiguous admission register/drain protocol proof

R10

Requirement / source: Per-attempt writer takeover fencing, V5 p7,10,50

Status: FAIL

Current implementation / evidence: Revision conflict handling is not a writer generation or explicit takeover lease

R11

Requirement / source: No structured key/solution fields in active-question responses, V3 p3; V5 p12

Status: PASS, scoped

Current implementation / evidence: `attempts.py:question_public` allowlists candidate fields and guards solution kind. This does not prove approved image crops contain no answer information

R12

Requirement / source: Offline acknowledged-vs-pending recovery, V2 p17; V5 p41–42

Status: PARTIAL

Current implementation / evidence: Local drafts, reconciliation and conflict UI exist; report28 shows client timer continues offline. Full reload/offline-open capsule and every malformed-storage case not verified

R13

Requirement / source: Safe local-draft lifecycle across profiles, V2 p17

Status: FAIL

Current implementation / evidence: F02: logout clears every profile's draft key without immediate discard confirmation. Privacy page documents logout removal but not a scoped preservation workflow

R14

Requirement / source: Defensive malformed storage recovery, V5 p41–42

Status: NOT VERIFIED

Current implementation / evidence: Parsed null reaches `Object.entries` in source. Conflicting browser probes prevent an actual UI pass/fail claim; language-only reproduction is insufficient

R15

Requirement / source: Server timer independent of analytics, V2 p16; V4 p8

Status: PARTIAL

Current implementation / evidence: Client timer uses server offset; API save guard and expiry exist. Full deadline/clock/lock failover qualification not performed

R16

Requirement / source: Durable submit receipt then async grading, V4 p1–2; V5 p31

Status: PARTIAL

Current implementation / evidence: Lightweight terminal update + grading marker +202pending UI exists; no full admitted-command watermark/drain protocol, and dispatcher/fencing defects remain

R17

Requirement / source: Accurate mathematical visual text, V3 p5–7,22

Status: FAIL

Current implementation / evidence: F03: typed MCQ options show raw LaTeX even though shared question math rendering exists

R18

Requirement / source: Complete accessibility/device/keyboard certification, V1 p12; V3 p22; V5 p42–43

Status: NOT VERIFIED

Current implementation / evidence: Label/focus/status patterns exist; no full assistive-tech/device-lab evidence. Main audit found two undersized mobile targets; PDF tagging is explicitly not claimed

R19

Requirement / source: Calm responsive navigation and honest UI states, V1 p4–8,11

Status: PARTIAL

Current implementation / evidence: Core routes and pending/error states exist; source-image/timed synthetic flows reviewed. Template browser title, mixed native controls, partial tokenization and repeated test IDs remain

Academic content, sourcing and protected publication

R20

Requirement / source: Source provenance, hashes and location-backed matching, V3 p1–8,16

Status: PARTIAL

Current implementation / evidence: Private upload metadata, hashes, q/s refs, crop checks and review flags exist. Full question-block/solution-block graph, cardinality forms, taxonomy and diff evidence are incomplete

R21

Requirement / source: Rights per audience/expiry fail closed, V3 p2–3; V5 p11–13

Status: FAIL

Current implementation / evidence: A04 invalid audience is accepted and fails open. Rights flags and restricted lists exist, but creation validation is inconsistent

R22

Requirement / source: Human semantic approval before public release, V3 p13–14,20

Status: PARTIAL

Current implementation / evidence: Approval flag + rights/accessibility checks exist; not independent role-separated approval or payload-bound attestations

R23

Requirement / source: Concurrent editorial revision integrity, V3 p13–14,20

Status: FAIL

Current implementation / evidence: A03 last-writer overwrites another reviewer edit without expected-revision conflict

R24

Requirement / source: One immutable academic publication identity per version, V5 p17–20,29

Status: FAIL

Current implementation / evidence: A02 concurrent publish produces duplicate same-version archives; archive uniqueness/atomic publication transition missing

R25

Requirement / source: Immutable attempt snapshots and grade lineage, V4 p2,21

Status: PARTIAL

Current implementation / evidence: Content-addressed packs and set-on-insert grade ledgers/correction variants exist; no signed lineage/WORM or complete semantic rule-bundle attestation

R26

Requirement / source: Four separately governed artifact classes, V3 p3; V5 p12,17–27

Status: FAIL

Current implementation / evidence: Source, candidate crops, server keys and embargoed solutions share application identity/compute path; release allowlists are not independent storage/workload boundaries

R27

Requirement / source: Cryptographically signed activation decisions and one-use grants, V5 p13–20,27

Status: FAIL

Current implementation / evidence: No full request/grant/result tuple, canonical payload digest, independent signing authority or consumed-grant tombstone workflow

R28

Requirement / source: Key custody, WORM/ObjectLock and storage policies, V5 p17–27

Status: NOT VERIFIED

Current implementation / evidence: No evidence of configured KMS/HSM custody or certified retention controls. Do not infer provider policy from a private object path

R29

Requirement / source: Hostile-document parser sandbox and budgets, V3 p7,23; V5 p39–40

Status: FAIL

Current implementation / evidence: Parsing occurs in live/shared deployment or threadpool. No production sandbox/network-denial/hard-kill isolation proof; S01 exposes a form-parser budget gap

R30

Requirement / source: Extraction output fencing, V5 p10,40,50

Status: FAIL

Current implementation / evidence: A06 check/write race lets a stale lease mutate the draft after authority changes

R31

Requirement / source: Original guide/source paper stays unapproved until reviewed

Status: PASS, scoped

Current implementation / evidence: Read-only DB snapshot: original75-question paper remains review, v1/r3; all50QAsets remain withdrawn. No audit publication performed

R32

Requirement / source: Authorized immutable content distribution and prefetch, V5 p47

Status: PARTIAL

Current implementation / evidence: Authenticated cropped assets/local byte-bounded cache exist; no complete publish-time candidate pack + authorization-aware CDN distribution protocol

Backend, database, operations and security

R33

Requirement / source: Modular ownership and API boundary, V1 p15; V5 p2,28

Status: PARTIAL

Current implementation / evidence: Domain modules and `/api` router boundary exist, but shared DB/runtime reads cross domains; this is not isolated service/cell ownership

R34

Requirement / source: Full bounded typed API catalogue, V5 p28

Status: PARTIAL

Current implementation / evidence: 46routes;2explicitresponsemodels;10dictbodyhandlers. Envelope version/requestID present; comprehensive typed input/output/error/idempotency conventions incomplete

R35

Requirement / source: Bounded JSON and ambiguous input rejection, V5 p28,52

Status: PASS, scoped

Current implementation / evidence: Actual JSON middleware caps1MiB/depth40 and rejects duplicate/nonfinite values; historical tests cover this. This row excludes forms/files

R36

Requirement / source: Multipart/form parsing limits, V5 p39–40,52

Status: FAIL

Current implementation / evidence: Corrected local ASGI report30 accepts2MiBnonfilefield with actual middleware; installed Starlette version has known relevant resource advisories

R37

Requirement / source: Credential security and privileged operations, V5 p9,13–16

Status: PARTIAL

Current implementation / evidence: Hashes, revocable sessions, staff throttle exist. MFA, one-use step-up, independent review and production secret lifecycle absent. Audit-process credential leak requires authorized rotation

R38

Requirement / source: Autonomous durable batch dispatch, V5 p31,46,52

Status: FAIL

Current implementation / evidence: A05 queue can remain idle after capacity frees; cron/poll-triggered dispatch is not a continuously serviced worker fleet

R39

Requirement / source: Idempotent output publication and worker write fences, V5 p10,40,52

Status: PARTIAL

Current implementation / evidence: Unique per-lease PDF path and job lease guards exist; extraction destination fence fails A06

R40

Requirement / source: Batch/live CPU/memory/identity failure isolation, V5 p44–46

Status: FAIL

Current implementation / evidence: Per-process lanes share deployment/threadpool/runtime; no independent worker/resource/trust plane

R41

Requirement / source: Bounded DB connections and waiting, V5 p49,52

Status: PARTIAL

Current implementation / evidence: Configurable default32connections/2swait. No fleet-wide pool budget, reserved critical pools or measured saturation evidence

R42

Requirement / source: Slim hot-path queries and budgets, V5 p49–50

Status: PARTIAL

Current implementation / evidence: Visit/history projections improved; per-start pack canonicalization/compression/upsert and repeated authenticated asset reads remain

R43

Requirement / source: Independent home cells and stable routing, V5 p44–45

Status: FAIL

Current implementation / evidence: No cell router, home-cell authority or independently provisioned writers/failure domains in application implementation

R44

Requirement / source: Atomic admission reservations and capacity release, V5 p7,30,50–51

Status: FAIL

Current implementation / evidence: No owner-bound preissued reservation consumption/counter transaction/exactly-once capacity release

R45

Requirement / source: Multi-AZ acknowledged durability and fenced DB failover, V5 p49–50

Status: NOT VERIFIED

Current implementation / evidence: Current accessible Mongo topology is standalone. Majority+journal request does not prove replicated durability or regional recovery

R46

Requirement / source: Degradation order, reserved terminal capacity and quota classes, V5 p51–52

Status: FAIL

Current implementation / evidence: Local limits and Retry-After exist, not admission/deadline-pressure controls that protect already-admitted cells

R47

Requirement / source: Cache correctness and nonauthority, V5 p47,49

Status: PARTIAL

Current implementation / evidence: Byte-bounded caches and authorization before crop delivery exist; distribution-tier authorization/invalidation/isolation is unqualified

R48

Requirement / source: WAF, edge quotas and hostile mixed traffic, V5 p52,64–68

Status: NOT VERIFIED

Current implementation / evidence: In-app limits are not evidence of external WAF policies or mixed authorized/hostile load qualification

R49

Requirement / source: Observability, bounded labels and attributable receipts, V5 p56–58

Status: PARTIAL

Current implementation / evidence: Audit events/request IDs/operational timestamp exist; no complete exported cell metrics, SLO/error-budget suite or incident evidence pipeline

R50

Requirement / source: Backups, restores and disaster recovery, V5 p53–55,64–68

Status: NOT VERIFIED

Current implementation / evidence: No operator-supplied restore/failover artifacts, production-volume timings, key-recovery rehearsal or RPO/RTO proof

R51

Requirement / source: 500k/2×/pathological workload qualification, V5 p3,46,48,64–68

Status: NOT VERIFIED

Current implementation / evidence: Capacity worksheet only; small8-way duplicate-command checks and isolated worker tests are not capacity qualification

R52

Requirement / source: Dependency/supply-chain gates, V5 p9,39,56–68

Status: FAIL

Current implementation / evidence: Current dependency scans report version-matched advisories; reachability/remediation not completed. No clean golden dependency gate demonstrated

R53

Requirement / source: Minimized telemetry and privacy policy, V4 p3–4; V5 p56

Status: PARTIAL

Current implementation / evidence: Explicit phone profile disclosure is good; public HTML enables product analytics/autocapture configuration needing reviewed production consent/minimization. No sensitive-export claim made

Results, analytics, reports and evidence governance

R54

Requirement / source: Exact arithmetic for supported scoring, V2 p18; V4 p2

Status: PASS, scoped

Current implementation / evidence: Actual scorer seven-case baseline passed. No claim of complete official adapter golden corpus

R55

Requirement / source: Regrade does not silently replace original score/report, V4 p2,21

Status: PARTIAL

Current implementation / evidence: Separate ledger/version objects and PDF paths exist; full independent release approval/correction-cohort policy incomplete

R56

Requirement / source: Complete matched-solution PDF, V3 p16,20; V4 p1,22

Status: PARTIAL

Current implementation / evidence: Full-report generator and access-controlled queued downloads exist, with historical real PDF checks. This audit did not repeat every75Q/render/accessibility variant or high-volume export campaign

R57

Requirement / source: All60analysis contracts and versioned unavailable states, V4 p1,5–16; V5 p63–64

Status: FAIL

Current implementation / evidence: `analytics_60_matrix.csv`:10related contracts partial,50notimplemented; no full PASS asserted. Internal40minimum is incorrect

R58

Requirement / source: Cohort definitions, privacy and sample thresholds, V4 p3–4,17–20

Status: FAIL

Current implementation / evidence: No full cohort snapshot/eligibility/suppression/percentile pipeline; absence of fake percentile claims is good but is not implementation

R59

Requirement / source: Question timing, confidence, revision/exposure evidence and action plans, V4 p8–16

Status: FAIL

Current implementation / evidence: No complete reconciled visibility ledger, confidence capture, canonical exposure/review history or7/30day planning engine

R60

Requirement / source: Test/golden/artifact registry sufficient for sign-off, V5 p57,64–68

Status: FAIL

Current implementation / evidence: Useful tests exist, but audit found copied simulations, no-op assertions, invalid selectors/encodings and scope leaks. Corrected narrow proofs do not replace the required complete evidence catalogue

R61

Requirement / source: Current build/compile health

Status: PASS, scoped

Current implementation / evidence: Frontend production build and Python compile pass; services run. This does not imply semantic correctness or deployment capacity

R62

Requirement / source: No application fixes during this audit

Status: PASS, scoped

Current implementation / evidence: Initial97sourcehashesunchanged; supplementaryfilesunchangedsincediscovery. Test-artifact leakage/cleanup and credential-report redaction disclosed separately

See `RULEBOOK_ERRATA.md` for internal documentation corrections and `AUDIT_REPORT.md` §9 for the proposed remediation order. None of these findings has been repaired during this audit.

Appendix E — All 60 analysis records

Source: docs/audit/analytics_60_matrix.csv. These are historical implementation-status records, not full metric definitions or passing acceptance tests. All 60 analyses are mandatory. Analytics IDs A01–A60 are a separate namespace from audit defect IDs A01–A08.

A01 — Net awarded marks

PDF-04 page: 5

Status: PARTIAL

Observation: Exact score exists; full metric card contract/drill-through metadata incomplete

Evidence: backend/tejas/scoring.py; frontend/src/features/Results.js; backend/tejas/corrections.py

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A02 — Fully correct response count

PDF-04 page: 5

Status: PARTIAL

Observation: Correct count and review filter exist for supported MCQ/integer custom practice

Evidence: backend/tejas/scoring.py; frontend/src/features/Results.js; backend/tejas/corrections.py

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A03 — Incorrect and partial outcomes

PDF-04 page: 5

Status: PARTIAL

Observation: Incorrect outcomes exist; partial-credit adapters and distinct partial analysis absent

Evidence: backend/tejas/scoring.py; frontend/src/features/Results.js; backend/tejas/corrections.py

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A04 — Attempted-answer accuracy

PDF-04 page: 5

Status: PARTIAL

Observation: Accuracy formula exists and baseline tested; full fraction/evidence contract incomplete

Evidence: backend/tejas/scoring.py; frontend/src/features/Results.js; backend/tejas/corrections.py

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A05 — Response completion rate

PDF-04 page: 5

Status: PARTIAL

Observation: Attempted/eligible counts and attempt_rate exist; complete metric workflow absent

Evidence: backend/tejas/scoring.py; frontend/src/features/Results.js; backend/tejas/corrections.py

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A06 — Ordinary positive marks earned

PDF-04 page: 6

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A07 — Penalty marks lost

PDF-04 page: 6

Status: PARTIAL

Observation: Penalty sum exists; full cross-adapter analysis contract incomplete

Evidence: backend/tejas/scoring.py; frontend/src/features/Results.js; backend/tejas/corrections.py

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A08 — Penalty-to-positive-marks ratio

PDF-04 page: 6

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A09 — Percentage of configured maximum marks

PDF-04 page: 6

Status: PARTIAL

Observation: Configured-maximum percentage exists; broader approved-adapter coverage absent

Evidence: backend/tejas/scoring.py; frontend/src/features/Results.js; backend/tejas/corrections.py

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A10 — Subject marks contribution

PDF-04 page: 6

Status: PARTIAL

Observation: Flat subject net sums exist; gross/penalty/award decomposition and taxonomy contract absent

Evidence: backend/tejas/scoring.py; frontend/src/features/Results.js; backend/tejas/corrections.py

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A11 — Chapter mark contribution

PDF-04 page: 7

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A12 — Concept attempted accuracy

PDF-04 page: 7

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A13 — Response-type performance

PDF-04 page: 7

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A14 — Editorial difficulty outcome matrix

PDF-04 page: 7

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A15 — First-exposure versus repeat performance

PDF-04 page: 7

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A16 — Timer time consumed

PDF-04 page: 8

Status: PARTIAL

Observation: Server elapsed total exists for simple custom timing; adjustments/visibility ledger absent

Evidence: backend/tejas/scoring.py; frontend/src/features/Results.js; backend/tejas/corrections.py

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A17 — Observed visible question time

PDF-04 page: 8

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A18 — Unobserved timer fraction

PDF-04 page: 8

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A19 — Time on slow incorrect responses

PDF-04 page: 8

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A20 — Question budget overrun

PDF-04 page: 8

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A21 — Subject time allocation

PDF-04 page: 9

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A22 — First-visit duration

PDF-04 page: 9

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A23 — Revisit time share

PDF-04 page: 9

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A24 — Positive score difference after revisions

PDF-04 page: 9

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A25 — Negative score difference after revisions

PDF-04 page: 9

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A26 — Late-window attempted accuracy

PDF-04 page: 10

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A27 — Recovery of unanswered review items

PDF-04 page: 10

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A28 — Unvisited eligible-item rate

PDF-04 page: 10

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A29 — Time spent on viewed omissions

PDF-04 page: 10

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A30 — Median time on fully correct items

PDF-04 page: 10

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A31 — Overall platform cohort percentile

PDF-04 page: 11

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A32 — Subject platform cohort percentile

PDF-04 page: 11

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A33 — Item full-correct cohort proportion

PDF-04 page: 11

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A34 — Item accuracy among attempted responses

PDF-04 page: 11

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A35 — Committed option-selection distribution

PDF-04 page: 11

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A36 — Correct-response time percentile

PDF-04 page: 12

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A37 — Cohort median correct-response time

PDF-04 page: 12

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A38 — Personal-to-benchmark time ratio

PDF-04 page: 12

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A39 — Item omission cohort rate

PDF-04 page: 12

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A40 — Item upper-minus-lower response difference

PDF-04 page: 12

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A41 — Confidence calibration by bin

PDF-04 page: 13

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A42 — Binary Brier score

PDF-04 page: 13

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A43 — Low-confidence penalty exposure

PDF-04 page: 13

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A44 — Assumption-based expected marks

PDF-04 page: 13

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A45 — Confirmed error-cause distribution

PDF-04 page: 13

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A46 — Comparable first-attempt score change

PDF-04 page: 14

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A47 — Fresh concept accuracy change

PDF-04 page: 14

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A48 — Comparable score variability

PDF-04 page: 14

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A49 — Repeated-item score difference

PDF-04 page: 14

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A50 — Unresolved review backlog

PDF-04 page: 14

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A51 — Recurring concept error count

PDF-04 page: 15

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A52 — Fresh review-check success rate

PDF-04 page: 15

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A53 — Delayed fresh-check accuracy

PDF-04 page: 15

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A54 — Assigned prerequisite coverage

PDF-04 page: 15

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A55 — Error-category marks-at-stake ceiling

PDF-04 page: 15

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A56 — Released-solution consultation rate

PDF-04 page: 16

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A57 — Observed solution review duration

PDF-04 page: 16

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A58 — Approved-key score revision impact

PDF-04 page: 16

Status: PARTIAL

Observation: Separate original/corrected scores exist; complete item-delta analysis not implemented

Evidence: backend/tejas/scoring.py; frontend/src/features/Results.js; backend/tejas/corrections.py

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A59 — Approved explanation availability

PDF-04 page: 16

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

A60 — Usable timing evidence coverage

PDF-04 page: 16

Status: FAIL_NOT_IMPLEMENTED

Observation: No corresponding versioned calculation + metric-specific unavailable-state contract found

Evidence: No matching metric registry/processor in inventoried application; underlying source data may exist but is not this analysis

Scope note: This grades the full PDF metric, not just a related scalar; no pass percentage or certified coverage claim

Appendix F — Traceability and source hashes

The following SHA-256 identities bind this export to its source snapshot. The full text of the allowlisted Markdown documents and original analytics CSV is also embedded as PDF file attachments for exact recovery in compatible PDF tools. Raw test reports, original specification PDFs, credentials and source URLs are excluded.

docs/strategy/CONTEXT_RESTORE_SNAPSHOT.md

```
SHA-256: 02c980ecf884f4f0e69193ba8d28d360139397aa4c78fe3fd87617623812b7b9
```

docs/strategy/TEN_PACKET_STRATEGY.md

```
SHA-256: b32251be1d9beefe71995fa2f3978f6e652f73b33cf361f2c31e4e1f9618f058
```

docs/strategy/SLO_AND_WORKLOAD.md

```
SHA-256: 5fa359623f033569e75f368d54aa23be4fd2a803f685c3e655c20c3ad392f9f2
```

docs/strategy/QUALIFICATION_AND_HANDOFF.md

```
SHA-256: 0b92942b23588a00b2cd995754f6af3690f19c9cb7fd8b0c8101b22997c2ce2a
```

docs/audit/AUDIT_REPORT.md

```
SHA-256: 274d9d55cef7a73d2c1c7f449b8af4cab151c58a68b8a4e7872bfafe6737b586
```

docs/audit/RULEBOOK_ERRATA.md

```
SHA-256: c84ec5c78953c1613fe4f3156011e426017bf832544674531f97511d1d35ee54
```

docs/audit/REQUIREMENTS_MATRIX.md

```
SHA-256: c4753d011417fcd31db9250ac2cf736e1ead59223d523ced61cf82378304612c
```

docs/audit/analytics_60_matrix.csv

```
SHA-256: 7926392e3996ab3d94ffba805f74b62062e8c8dcafa5429d56cd807bb9a7304d
```

Source-bundle digest (canonical source inventory):
635eae0ff5b3f58b856a430c8761e67bac828592e66a8f8c86c7c7713b609c8e

Generator: docs/strategy/export_strategy_pdf.py

```
SHA-256: c525de24d1cf21a8ee7813c07cd59d656b8c696158797bed6c6e87a762f6c493
```

Renderer dependencies: ReportLab 4.4.10; Markdown 3.10.2; PyMuPDF 1.27.1

Repository baseline commit: e1f76fc363d6d3724cf97c41178d5fd8a32fab8e. The working tree also contains uncommitted documentation; the file hashes above identify the exported content.

Original five specification PDFs — identities only

01_Product_Blue_Design_and_Master_Prompt.pdf

Pages: 27. Original PDF not embedded.

```
SHA-256: c672e1188184745866cc8c286b8f55afb28ea11651b98d4b8e6968687b695bc5
```

02_Exam_Formats_Scoring_and_Session_Engine.pdf

Pages: 25. Original PDF not embedded.

```
SHA-256: 4b4bcce2ef15704f47fc80929724cd9119baaa09bfbfccf898627024ec51434e
```

03_PDF_Questions_Solutions_and_Admin_Workflow.pdf

Pages: 23. Original PDF not embedded.

```
SHA-256: b5460afe0b9c7cf6a7bda8c76b2a40499555bcf4a728520499641ce7f7d84ecd
```

04_Results_60_Analyses_and_Question_Insights.pdf

Pages: 23. Original PDF not embedded.

```
SHA-256: cbf05a521cb329ebf6368e4c44b5e771f7d99bf1305633ec41e64c6ef7e0f765
```

05_Architecture_Scale_Security_and_Launch.pdf

Pages: 79. Original PDF not embedded.

```
SHA-256: 8e314d6d09be286f8a94774a09b97626efada49b57128486080159e1fa3bd5b0
```

Total: 177 source pages. This identity list does not replace the original clauses or prove complete atomic traceability.

Appendix G — Status and next development action

All ten packets: NOT STARTED implementation at this snapshot. Packet 1 is approved next; later packets remain dependency-gated.

Open correctness/security findings are recorded in Appendix C; their owning repair packets are listed in Appendix B. No application code, database, environment value, credential or session was changed to produce this export.

Credential incident: rotation and session revocation are still pending separate confirmation and secure authorized execution.

The original paper and QA fixture status in the audit are historical observations, not freshly verified database state in this export.

The strategy references two absent coverage artifacts: coverage_manifest.json and COVERAGE_MATRIX.md. Resolve this in Packet 1; do not assume the 62-row audit matrix maps every mandatory clause of all five PDFs.

Legacy rulebook guidance saying 40 analyses is superseded by the source requirement for all 60. The source strategy's "proposed for approval" label is historical; see the current user decision in Context restore. No approval of all proposed SLOs, operator decisions or launch readiness is inferred.

Next: inspect the current repository, then execute Packet 1's rules, safety and regression baseline. Do not silently implement later packets or claim 500k readiness.

This is a searchable, bookmarked reference PDF. Formal tagged-PDF/PDF-UA accessibility certification is not claimed.